

02-741 Week 1

Generative AI for Biomedicine

Aidan Jan

August 27, 2026

Introduction to Molecular and Cell Biology

We don't quite understand ourselves. In the human body, there are tens of trillions of cells, and dozens of different cell types. (Just neurons have over ten different types.) However, we do not understand how cells differentiate. For example, why a muscle cell originally grew into a muscle cell, or why a neuron originally grew into a neuron. Additionally, we don't know how to change one cell type into another cell type. (e.g., Skin cell $\rightarrow$ Stem cell $\rightarrow$ Neuron.)

One domain of current research is with gene expression. Since all cells have the same DNA, what features (or activation of genes) would cause a stem cell to become a certain cell type? Given some information on which genes are activated, can we predict what cell type the cell is? This is where AI comes in.

Decoding the “Language” of Genomes, Cells, and Tissues

Biology is multi-scale. We can observe gene sequences, cell structure, and cell function, all of which are at different scales. As a result, our models must also be multi-scale.

Central Dogma of Biology

Proteins do most of the work in biology, and are encoded by subsequences of DNA, known as genes.



The Human Genome: the “Blueprint” of Our Body

- 10^{13} different cells in an adult human
- The cell is the basic unit of life
- DNA = linear molecule inside the cell that carries instructions needed throughout the cell's life - long string(s) over a small alphabet
- Alphabet of four nucleotides / bases. (A, C, T, G)

DNA organizes into chromatin, which form into nucleosomes, which condenses into chromosomes. Humans have 22 pairs of chromosomes, plus the two sex chromosomes.

Genomes

- A genome is an organism's complete set of DNA (including its genes)
- In humans, less than 2% of the genome actually encodes for genes

FILL

In recent years (2001 - Present), high throughput DNA sequencing allows for cheap sequencing of human DNA. In 2001, one genome costs over \$100,000,000. Now (2026), it costs less than \$800 per genome. High-throughput DNA sequencing is one of the most disruptive technologies in the past decade.

Understanding the Genome

- View from 2000
 - Protein coding regions: 35000-120000
 - Regulatory sequence: Less than protein-coding information
 - Transposons: Junk DNA
 - (All of these assumptions are wrong!)
- We now know...
 - Protein coding regions: 20000. (Fewer than fish!)
 - Regulatory sequences: More regulatory sequences than protein-coding regions
 - Transposons: Make up half of the human genome
 - Gene numbers do not correlate with organism complexity
 - 5% of genes are conserved between different species, but only 1.5% actually codes for proteins

0.0.1 The Spectrum of Genetic Variation

[FILL 21]

Sequence Modeling and Generation with Transformer

Can we use GenAI to design molecules with desired functions?

- Medicine
- Vaccines
- Enzymes (Biocatalysts)
- Biosensors (e.g., GFP)
- New materials

Molecule generation is somewhat similar to language generation.

- Both are sequences of discrete tokens. (In language it's words, in molecules, it is atoms or amino acids.)
- Both have discrete structures. (In language it's grammar, in molecules it's chemistry)
- Geometry (Unique for molecules)

For proteins, they can be roughly represented as sequences of amino acids.

- Proteins are building blocks of life
- Responsible for important biological functions
- Built by sequences of amino acids.
 - GFP (Green Fluorescent Protein) = VLLPDNHYLSTQSALSKDPNEKRDHMLLEFVTAAGIT

Language Models

A language model is a probabilistic model of discrete sequences, including human languages and biological languages. e.g.,

Probability("Pittsburgh is a city of bridges")

Probability("VLLPDNHYLSTQSALSKDPNEKRDHMLLEFVTAAGIT")

Each next token can be defined as a probability based on previous tokens.

$$P(\text{next word } y_t | \text{Prompt } x, \text{previous words } y_{1:t-1})$$

For example,

- Santa Barbara has very nice _____
 - "beach" = 0.5
 - "weather" = 0.4
 - "snow" = 0.01
- Pittsburgh is a city of _____
 - "bridges" = 0.6
 - "corn" = 0.02

To calculate the probabilities, the probability of each word (or each amino acid) depends on all the previous tokens.

$$\begin{aligned} &P(\text{"Pittsburgh is a city of bridges"}) \\ &= P(\text{"Pittsburgh"}) \cdot P(\text{"is"} | \text{"Pittsburgh"}) \cdot P(\text{"a"} | \text{"Pittsburgh is"}) \cdot P(\text{"city"} | \dots) \\ &\cdot P(\text{"of"} | \dots) \cdot P(\text{"bridges"} | \dots) \end{aligned}$$

Or simply,

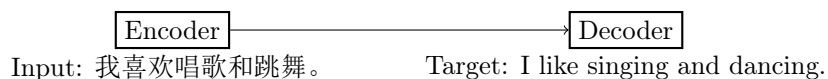
$$P(x_{1:T}) = \prod_{t=1}^T P(x_{t+1} | x_{1:t})$$

We can predict the $P(x_{t+1} | x_{1:t})$ part using neural networks (e.g., CNNs, RNNs)

Types of Language Models

- **Encoder-only Model** (or Masked LM, Non-autoregressive)
 - Examples: BERT, RoBERTa, ESM (for protein)
- **Encoder-decoder**
 - Example: T5
- **Decoder-only** (Autoregressive)
 - Examples: GPT, LLaMA, ProGen (for proteins)

Encoder-Decoder Paradigm

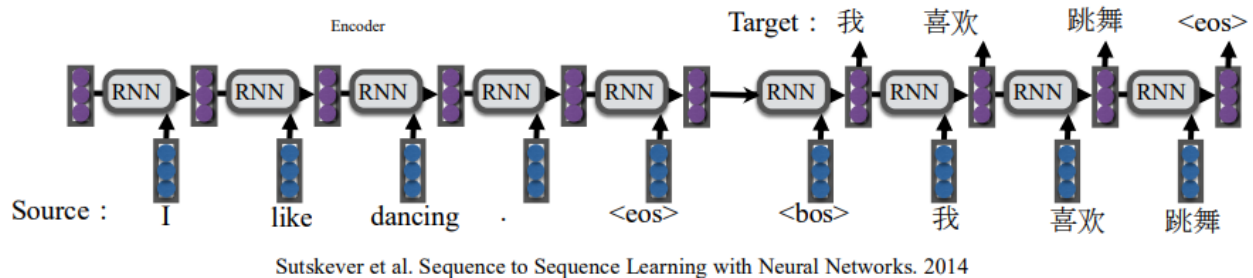


Sequence to Sequence Learning

- Conditional text generation: directly learning a function mapping from source sequence to target sequence

$$p_{\theta}(y|x) = \prod_t p(y_t|x, y_{1:t-1}; \theta)$$

- Previous encoder / decoder: LSTM or GRU



We use the encoder-decoder architecture because it is

- Full context and parallel: use Attention in both encoder and decoder
- Not recurrent $\Rightarrow$ concurrent encoding

Motivation of Attention: Translation between Different Alignment of Words



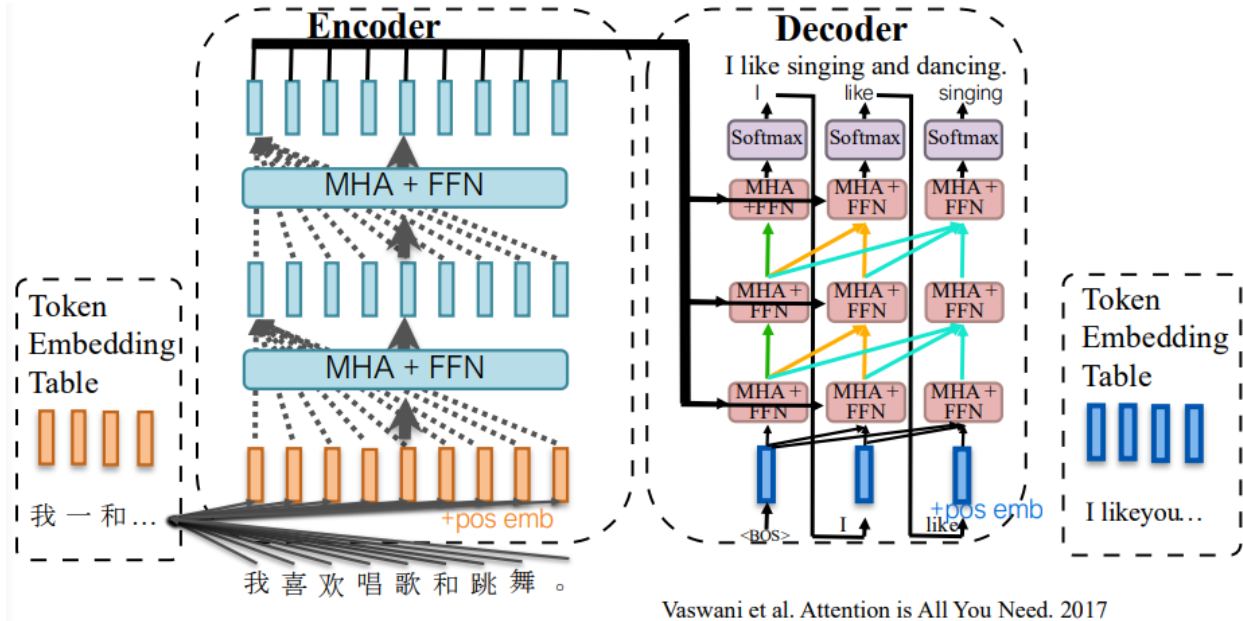
Different languages have different orderings for words.

Transformer-based Models for Biomolecules

- Protein Language Models:
 - ESM, ESM-Fold
 - ProGen
 - AlphaFold (with EvoFormer)
- Genomics
 - DNABERT / DNABERT2

- Evo, Evo2
- ENABERT
- Single Cell: scGPT, scFoundation

Transformers

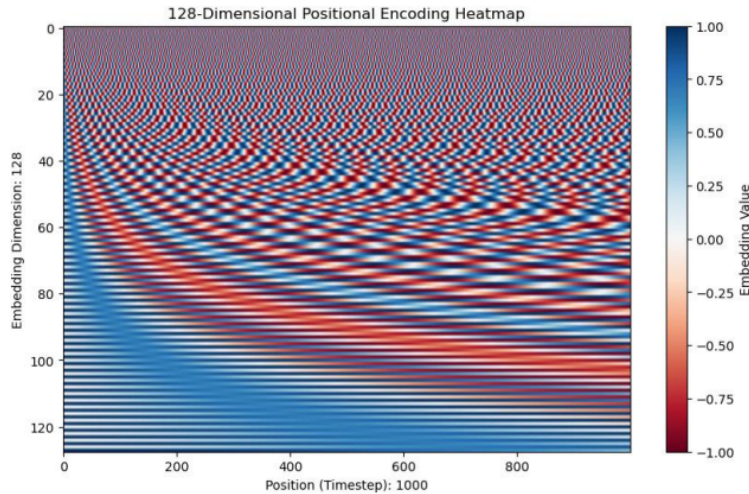


What is Embedding?

- Token Embedding:
 - token index $\rightarrow$ vector
 - Shared (tied) input and output embedding from lookup table
- Positional Embedding
 - to distinguish words in different positions, map position labels to embedding, dimension is same as token embedding
 - For t -th position, i -th dimension, the equation is:

$$PE_{t,2i} = \sin\left(\frac{t}{1000^{2i/d}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{1000^{2i/d}}\right)$$

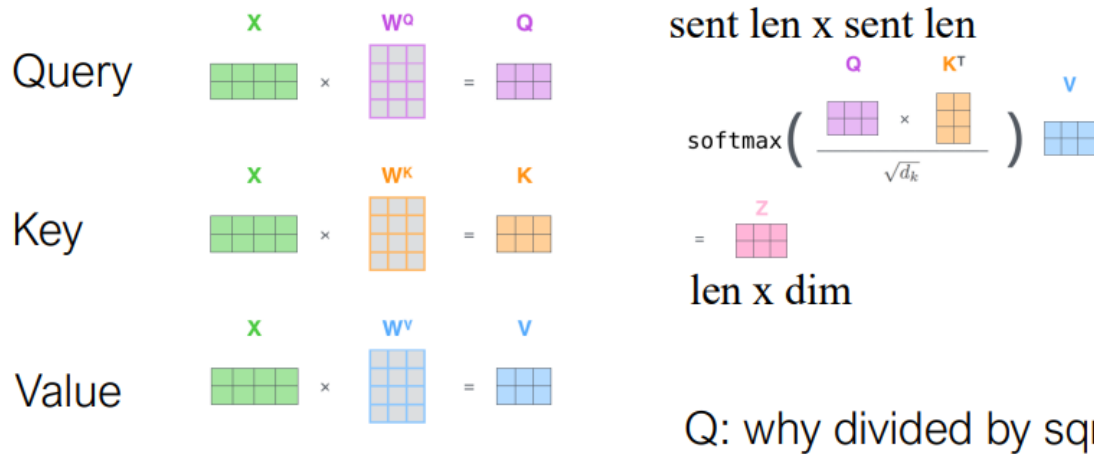


Multi-head Attention

- Instead of one vector for each token, we break it into multiple heads
- Each head performs attention.

$$\begin{aligned}\text{Head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \text{MultiHead}(Q, K, V) \\ &= \text{Concat}(\text{Head}_1, \text{Head}_2, \dots, \text{Head}_h)W^O\end{aligned}$$

X are input embeddings from previous layer (num of tok * dim)

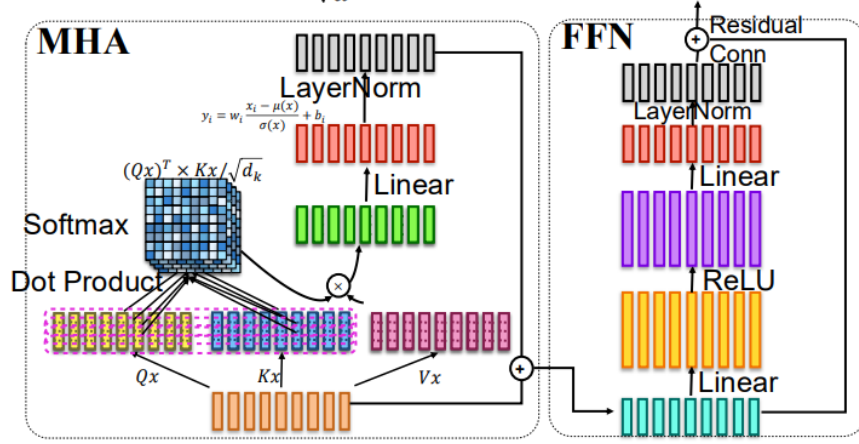


The $\sqrt{d_k}$ is a regularization coefficient, where d_k is the dimensionality of the attention (number of heads). This coefficient makes it softmax is a range from 0 to 1.

Multihead Attention and FFN

$$\text{Attention}(Q, K, V, x) = \text{Softmax}\left(\frac{(Qx)^T Kx}{\sqrt{d}}\right) \cdot (Vx)^T$$

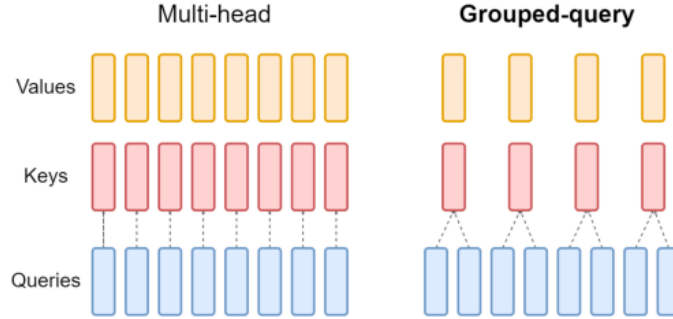
$$\text{FFN}(x) = \max(0, x \cdot W_1 + b_1) \cdot W_2 + b_2$$



Attention Variant: Grouped-Query Attention (GQA)

$$\text{Head}_i = \text{Attention}(XW_h^Q, XW_{g(h)}^K, XW_{g(h)}^V)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{Head}_1, \text{Head}_2, \dots, \text{Head}_h)W^O$$



GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints. Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, Sumit Sanghai. EMNLP 2023.

Grouping the queries can speed up the inference since there are less overall matrix multiplications.

Attention Variant: Multi-head Latent Attention (MLA)

- Keys and values occupy large GPU memory
 $- 2 \cdot \text{len} \cdot n_h \cdot d_h$
- instead of caching d_h vector (for each t_k , head), reduce to a latent space with rank r .

$$\begin{aligned} Q &= X \cdot W_Q^D \cdot W_Q^U \\ K &= X \cdot W_{KV}^D \cdot W_K^U \\ V &= X \cdot W_{KV}^D \cdot W_V^U \end{aligned}$$

The diagram shows the equations for Q, K, and V in MLA. Red arrows indicate the dimensions of the matrices: $d \times r$ for W_Q^D , $r \times (n \cdot d_h)$ for W_Q^U , and $r \times (n \cdot d_h)$ for W_K^U and W_V^U .

- First proposed in Deepseek-V2, used in all modern LLMs

- Key and Value share down-projection, and across all heads

$$\begin{aligned}
& MLA(x)_h \\
&= \text{Softmax} \left(\frac{X \cdot W_Q^D \cdot W_{Q,h}^U \cdot \left(X \cdot W_{KV}^D \cdot W_{K,h}^U \right)^T}{\sqrt{d_h}} \right) \cdot X \cdot W_{KV}^D \cdot W_{V,h}^U \\
&= \text{Softmax} \left(\frac{C_Q \cdot W_{QK,h} \cdot C_{KV}^T}{\sqrt{d_h}} \right) \cdot C_{KV} \cdot W_{V,h}^U
\end{aligned}$$

We can precompute and cache the following:

$$\begin{aligned}
W_{QK,h} &= W_{Q,h}^U \cdot (W_{K,h}^U)^T \\
C_Q &= X \cdot W_Q^D \\
C_{KV} &= X \cdot W_{KV}^D
\end{aligned}$$

Decoder Self-Attention Mask

- Encoder uses bidirectional attention
- Decoder uses unidirectional attention
 - Maskout right side before softmax (-inf)

FFN with SwiGLU

[Fill]